

Interoperability Is Not the Same as Longitudinal History

Why Healthcare AI Requires More Than TEFCA

The national longitudinal healthcare information architecture doesn't replace today's healthcare information systems. It enriches them by allowing each system to focus on what it does best while sharing capabilities that are better developed once and used by all.

This article builds upon two earlier papers. The first described a vision for using AI to reduce healthcare costs while improving patient care, accompanied by a proposal for a focused 60-day feasibility study. The second argued that AI isn't the architecture—it is one component of a broader information architecture. Before AI can reason effectively, it needs a trustworthy, evidence-linked longitudinal patient history.

This article steps back one layer in that architecture to examine where that history comes from and how existing healthcare information systems and TEFCA fit into the solution.

Health Care Information Systems

Modern healthcare information systems have transformed the way healthcare organizations operate. They support scheduling, registration, physician documentation, laboratory processing, imaging, pharmacy, billing, claims, and countless other operational processes that allow hospitals and physician practices to function efficiently.

At the same time, another transformation has occurred. The volume and diversity of clinical information generated about each patient has grown dramatically.

Today's healthcare generates an unprecedented volume of information, including:

- laboratory results
- imaging studies
- physician notes
- specialist consultations
- prescriptions
- wearable devices
- home monitoring
- genomics
- patient-generated information

Healthcare doesn't have a shortage of information. It has a shortage of integrated, trustworthy patient history.

A typical patient may receive care from a primary care physician, multiple specialists, hospitals, imaging centers, laboratories, pharmacies, and other providers. As people age, relocate, or change insurance, their medical history becomes distributed across an increasing number of healthcare information systems. As a result, no single organization typically possesses a complete longitudinal view of the patient's health history.

The challenge is no longer simply collecting healthcare information. It is transforming information scattered across many organizations into a trustworthy longitudinal patient history, then helping physicians quickly identify the small portion of that history that matters for today's clinical decision.

This is precisely the problem that existing healthcare information systems, interoperability standards such as TEFCA, and a national longitudinal patient history should work together to solve. Each has a different role within the overall information architecture.

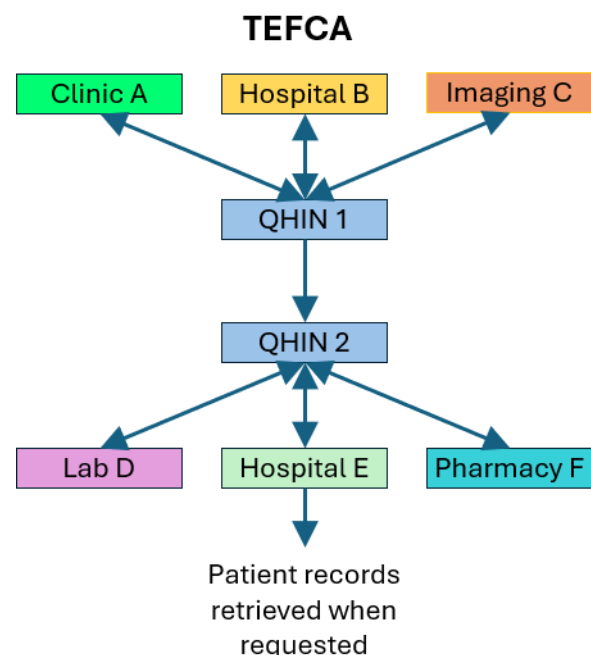
TEFCA: A Foundation for Healthcare Interoperability

Healthcare providers have long struggled to exchange patient information across organizational boundaries. Every hospital, physician practice, laboratory, imaging center, and pharmacy typically maintains its own healthcare information system. When patients receive care from multiple organizations, those systems must be able to locate and exchange relevant information while protecting patient privacy.

The **Trusted Exchange Framework and Common Agreement (TEFCA)** establishes a nationwide framework for healthcare interoperability. Rather than requiring every healthcare organization to create separate interfaces with every other organization, TEFCA defines common policies, technical standards, and trusted networks that allow participating organizations to discover and retrieve patient information from one another.

At a high level, healthcare organizations connect to one of several **Qualified Health Information Networks (QHINs)**. Those QHINs communicate with one another, allowing an authorized provider to locate and retrieve information about a patient from participating healthcare organizations across the country.

TEFCA represents a significant milestone in healthcare interoperability. It addresses one of



the industry's most difficult challenges: allowing authorized healthcare providers to discover where patient information exists and retrieve it using common standards rather than hundreds of proprietary interfaces.

Interoperability Is Not the Same as Longitudinal History

While TEFCA provides an essential interoperability framework, its purpose is fundamentally different from maintaining a continuously updated longitudinal patient history.

When a provider needs additional information about a patient, TEFCA helps locate participating organizations that may hold relevant records and retrieves those records on demand. It was not designed to continuously reconcile information arriving from multiple organizations, preserve evidence and provenance, maintain temporal history, or create a single trusted longitudinal patient history.

Those responsibilities belong to a different architectural layer.

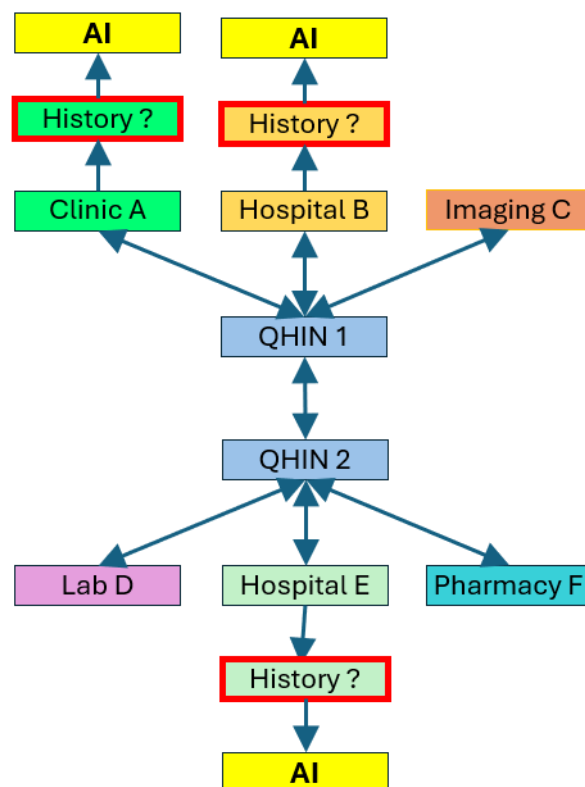
The Missing Architectural Layer

A natural question is whether every healthcare information system could simply retrieve patient records through TEFCA and perform its own clinical knowledge extraction, reconciliation, longitudinal history management, and AI reasoning. Technically, it could.

The architectural question is different:
Should every healthcare information system independently solve the same national problem?

There is another important consideration. Today's interoperability framework is designed primarily for **on-demand retrieval** of patient information. When new clinical information is created—or existing information is corrected—there is currently no nationwide mechanism that continuously notifies every

Independent Longitudinal History



How are these histories kept synchronized?

healthcare information system responsible for maintaining its own longitudinal history. Without such a capability, each system would either have to repeatedly query TEFCA looking for changes or risk maintaining an incomplete or outdated longitudinal history.

A national longitudinal architecture changes that model. Rather than requiring hundreds of healthcare information systems to independently discover, retrieve, interpret, reconcile, and maintain the same patient history, participating systems would publish new and corrected clinical events through a robust change-data-capture interface. The longitudinal architecture would continuously transform those events into a trusted, evidence-linked patient history while preserving provenance, temporal history, and supporting clinical evidence.

That approach allows healthcare information systems to continue focusing on the functions they perform exceptionally well: scheduling, physician workflow, documentation, laboratory processing, imaging, billing, claims, and patient interaction. At the same time, they gain access to a continuously maintained longitudinal patient history and shared AI services that would otherwise need to be developed and maintained independently by every vendor.

The National Longitudinal Healthcare Architecture

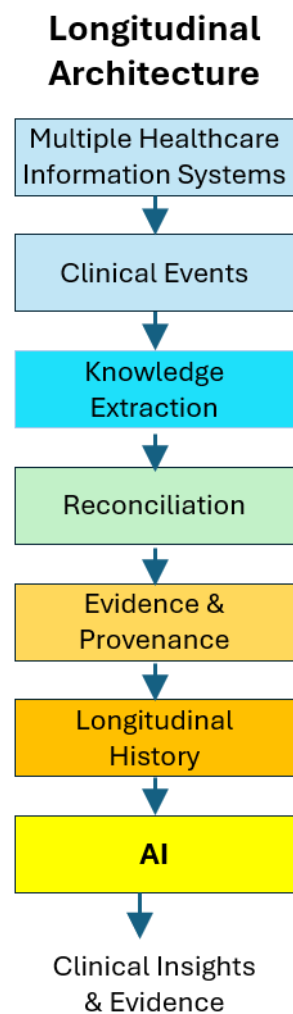
Rather than requiring every healthcare information system to independently discover changes, reconcile conflicting information, maintain its own longitudinal patient history, and develop AI capabilities, these responsibilities become shared services within a national information architecture.

As participating healthcare information systems create new clinical information or correct existing records, those clinical events are published to the longitudinal architecture. Structured clinical information can be processed directly, while clinical notes, imaging reports, and other unstructured information undergo clinical knowledge extraction to identify diagnoses, observations, procedures, medications, and other clinically relevant facts together with the supporting evidence from which they were derived.

Information arriving from multiple organizations is then reconciled. Rather than overwriting conflicting information, the architecture preserves provenance, supporting evidence, temporal history, and differing clinical assertions so that the longitudinal history reflects both the patient's current status and the evolution of that history over time.

The resulting longitudinal patient history becomes the foundation for AI-assisted clinical reasoning. Because AI is now operating on an integrated, evidence-linked patient history rather than isolated records retrieved during an encounter, it can identify trends, inconsistencies, medication interactions, unresolved issues, and other observations that may warrant discussion between the physician and patient. The architecture produces clinical insights supported by evidence rather than replacing clinical judgment.

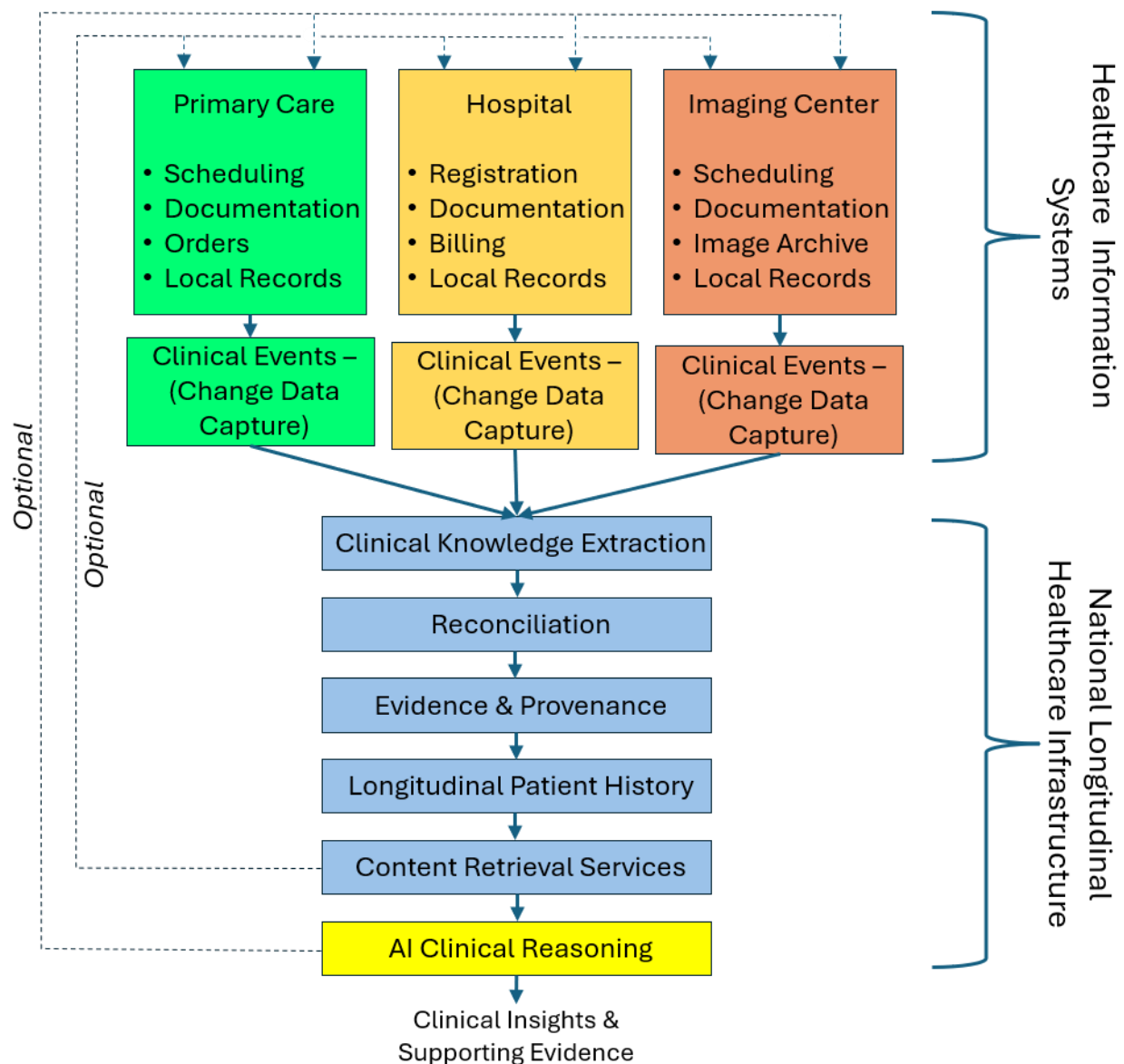
Healthcare information systems remain the systems of record for the clinical information they create. The national longitudinal architecture does not replace those systems; it continuously assembles, reconciles, and enriches their information into a trusted patient history that every participating healthcare organization can share.



Impact on Healthcare Information Systems

A national longitudinal healthcare architecture is intended to complement—not replace—today's healthcare information systems. Participating systems remain responsible for the operational functions that support hospitals, physician practices, laboratories, imaging centers, pharmacies, and payers. The national infrastructure assumes responsibility only for those capabilities that are inherently shared across organizations: assembling, reconciling, and maintaining a trustworthy longitudinal patient history while making that history available through common retrieval and AI services.

Division of Responsibility Within a National Healthcare Architecture



Participating healthcare information systems would continue to create and maintain the clinical records required to operate physician practices, hospitals, laboratories, imaging centers, pharmacies, and payer organizations. They remain the authoritative systems of record for the clinical information they generate.

The national longitudinal healthcare infrastructure becomes responsible for maintaining the integrated, evidence-linked longitudinal patient history, while participating healthcare information systems remain responsible for maintaining the operational records required to deliver care. This clear division of responsibility allows each layer of the architecture to focus on the functions it performs best.

To support a continuously maintained longitudinal patient history, participating systems would provide a robust Change Data Capture interface capable of making newly created, corrected, and deleted clinical information available to the national longitudinal healthcare infrastructure. The proposed interfaces build upon the interoperability standards established by TEFCA rather than replacing them. The primary addition is the ability to communicate changes as clinical information is created or revised, allowing the longitudinal patient history to remain continuously synchronized with participating healthcare information systems.

Supporting continuous synchronization requires more than simply transmitting clinical data. Clinical events must include persistent identifiers so that subsequent corrections or revisions can be associated with the original event rather than treated as entirely new information. Similar persistent identifiers may also be required for evolving collections of clinical information such as allergies, medication lists, patient permissions, implanted devices, and other clinical assertions whose contents change over time.

Once the longitudinal patient history has been assembled, healthcare information systems have flexibility in how they use it. Some organizations may retrieve relevant portions of the patient's history using standardized Content Retrieval Services. Others may also leverage nationally provided AI Clinical Reasoning services to identify trends, conflicting information, medication interactions, unresolved follow-up, and other topics deserving further clinical review. Organizations wishing to differentiate their products remain free to develop their own AI capabilities using the longitudinal patient history as a trusted foundation.

Perhaps the greatest benefit to healthcare information system vendors is that they no longer need to independently develop and maintain many of the complex capabilities associated with longitudinal history management. Clinical knowledge extraction, reconciliation of conflicting information, provenance management, temporal history,

maintenance of a trusted longitudinal patient history, and shared AI services become national infrastructure rather than capabilities repeatedly implemented by every vendor. Healthcare information systems remain free to focus on the areas where they provide the greatest value: intuitive user interfaces, efficient clinical workflows, scheduling, documentation, billing, practice management, and innovations that directly support healthcare providers and patients.

What This Could Look Like

Imagine arriving for your annual physical ten years from now.

Instead of completing several pages of forms listing surgeries, medications, specialists, allergies, and family history that have been entered countless times before, your physician's healthcare information system has already retrieved a trustworthy longitudinal patient history assembled from every participating healthcare organization that has cared for you.

Overnight, new laboratory results, imaging studies, specialist consultations, medication changes, and hospital visits have already been incorporated into that history. Conflicting information has been reconciled while preserving the original evidence and its source. Significant trends have been identified. Questions that deserve discussion have already been highlighted.

Before you enter the examination room, your physician receives a concise briefing rather than hundreds of pages of records.

"The patient's LDL cholesterol has increased steadily over the past four years despite medication."

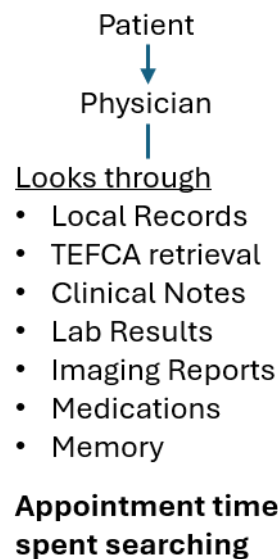
"Blood pressure has remained well controlled."

"A pulmonary nodule first identified three years ago has remained stable across four imaging studies."

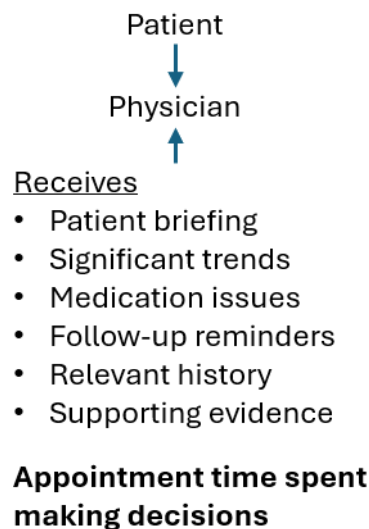
"A medication prescribed by one specialist may be contributing to symptoms reported to another."

"Family history suggests additional screening may be appropriate."

Today



Future



Rather than spending valuable appointment time searching for information or reconstructing years of medical history, physicians and patients begin the visit discussing the issues that matter most.

That is the opportunity.

The objective is not to replace physicians, healthcare information systems, or TEFCA. It is to build upon them by creating a national information architecture that transforms fragmented records into a trusted longitudinal patient history—allowing healthcare professionals to spend less time finding information and more time caring for patients.